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A Physics-guided Implicit Neural Representation for Streak Reduction in X-ray Dark-field CT

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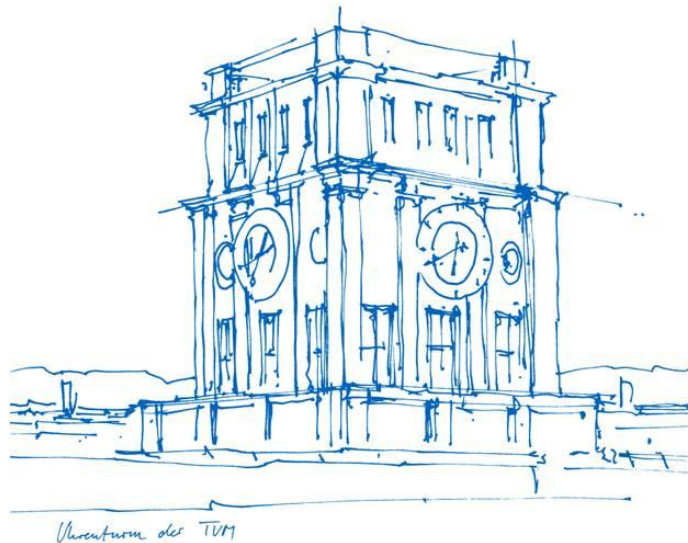
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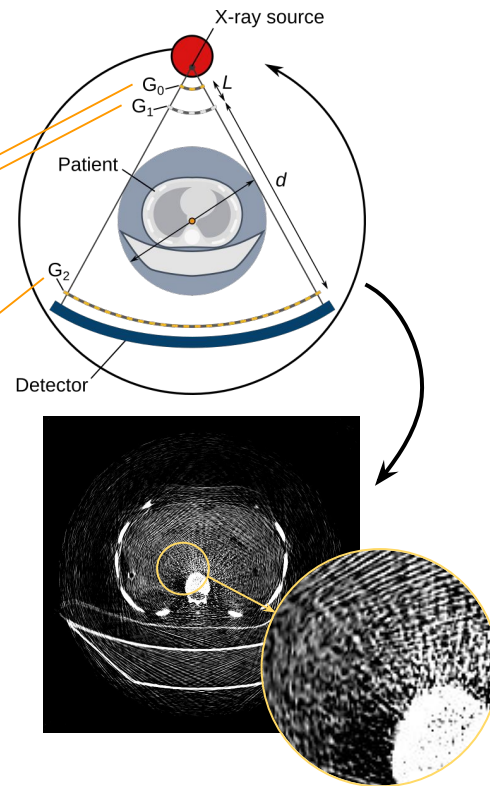
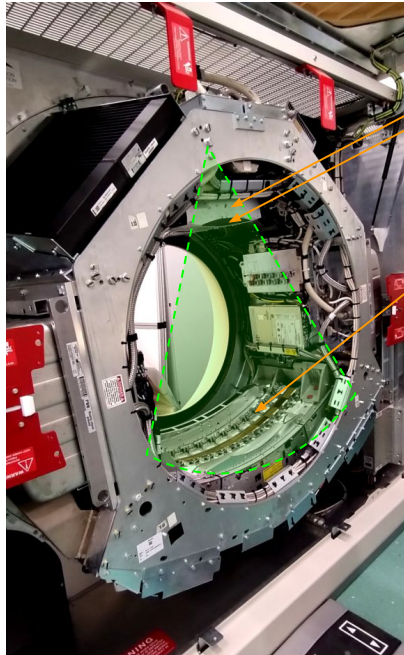
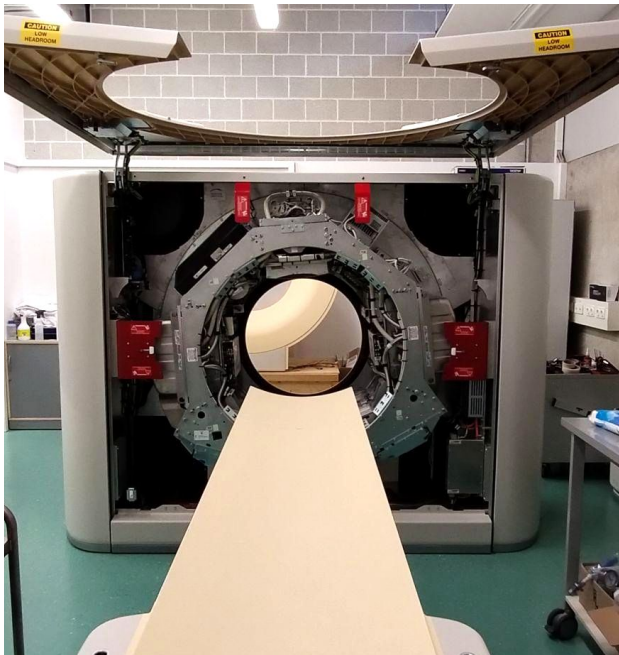
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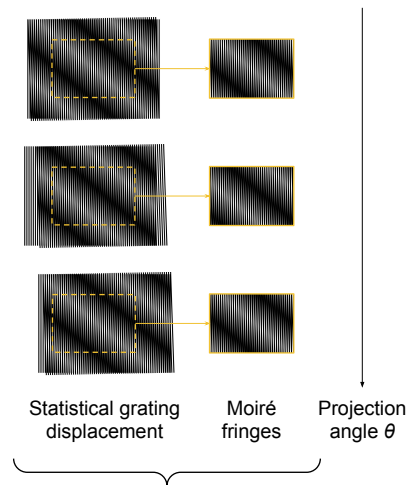


Background: Human-scale Dark-field CT Prototype

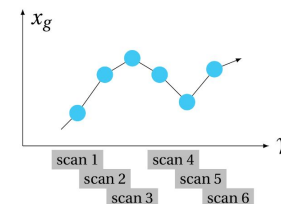


Philips iCT gantry operated at 550 mA, 1.5 s (825 mAs), 80 kVp

The Physics: Talbot–Lau Interferometry

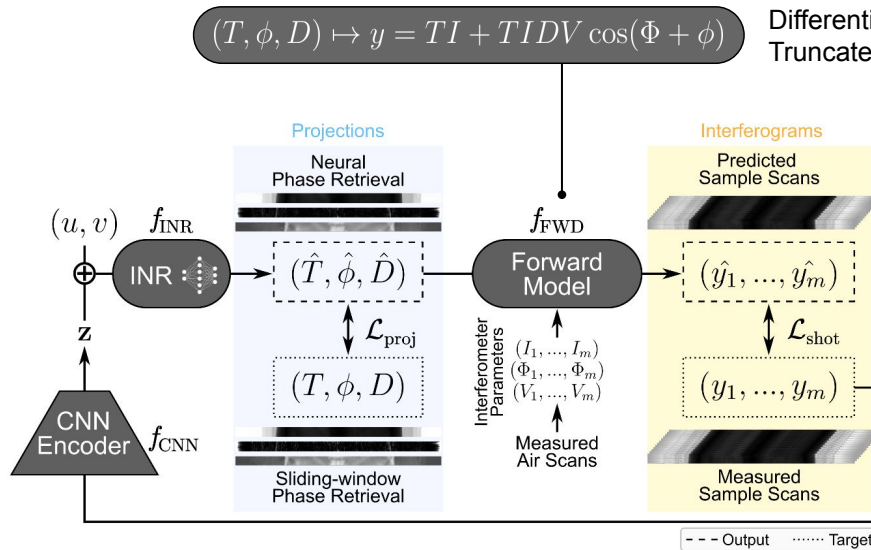


Continuous acquisition:
Sliding-window phase retrieval

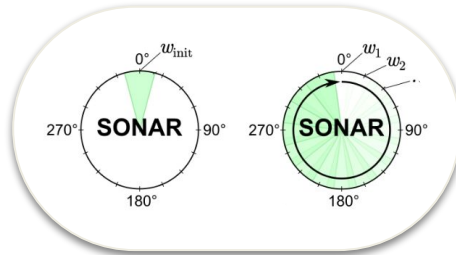


The Representation: Continuous Neural Field

SONAR “Shot-Optimized Neural Adaptive Representation”



Training: Warm-start fine-tuning
Initialization shot: 1000 iterations
Shot-wise fine-tuning: 50 iterations



Projections

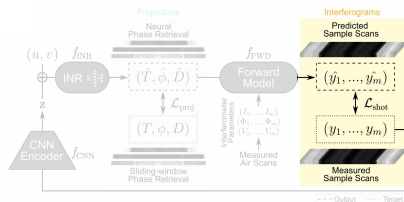


Forward Model
= Measurement

Phase Retrieval

Interferograms (or shots)





Shots (2400 across the full CT rotation)

Conventional

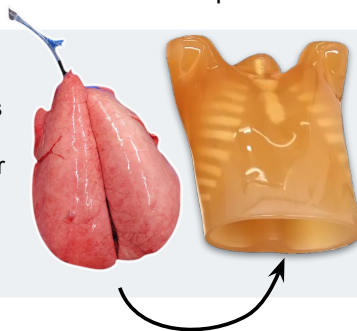


SONAR



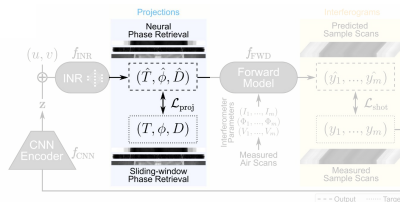
Measured sample:

Ex vivo
porcine lungs
Ventilated at
10 to 30 mbar

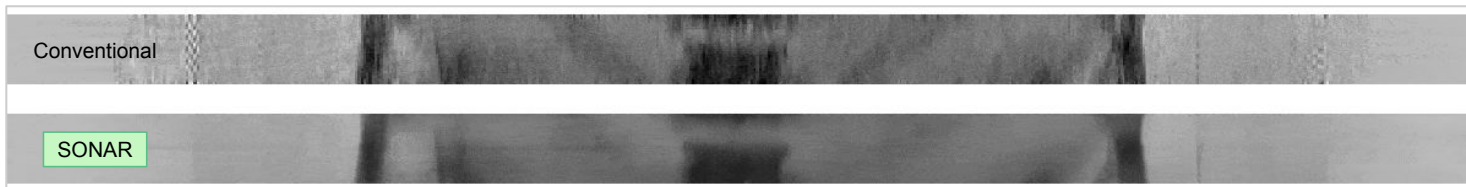


Human thorax
phantom
"Lungman"

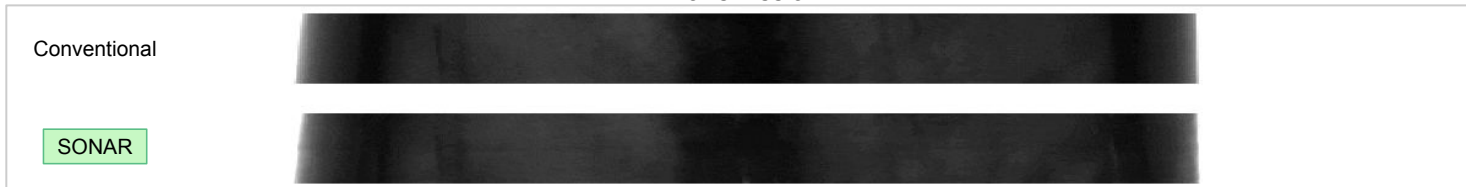
Projection Space



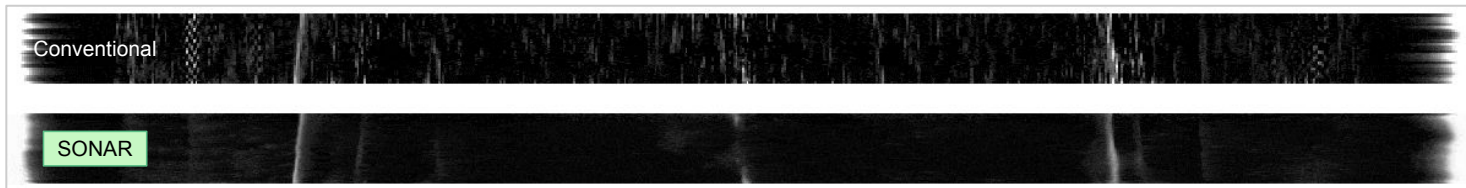
Dark-field



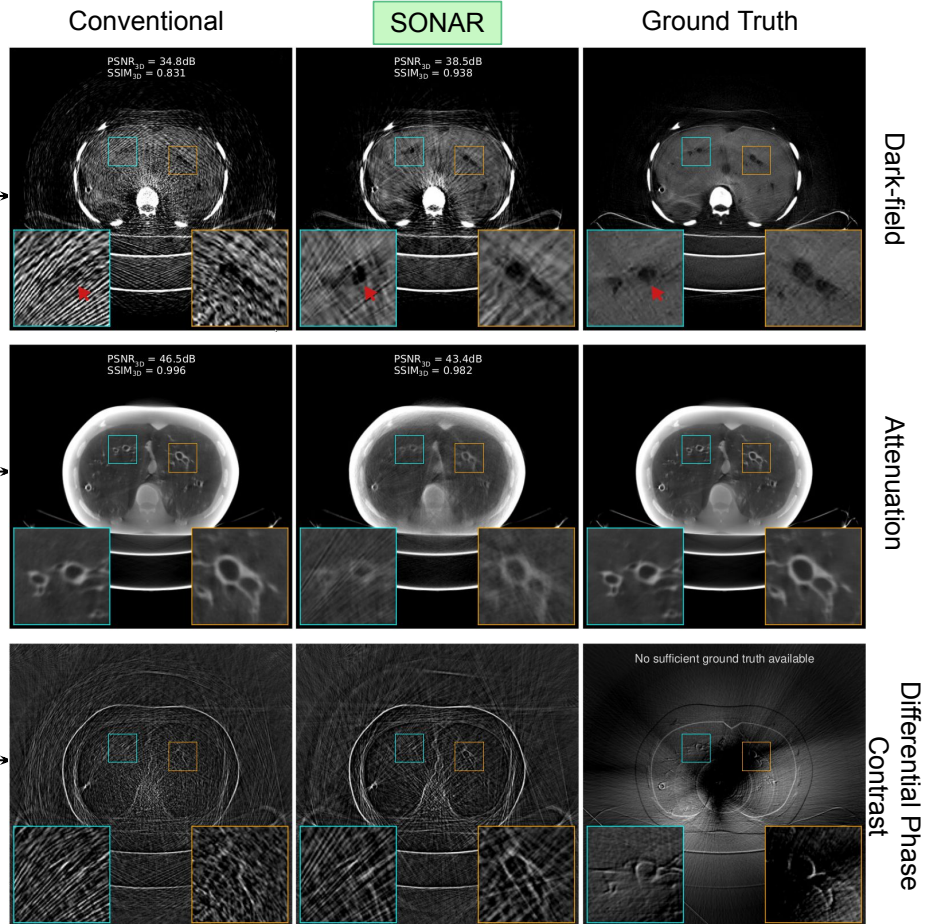
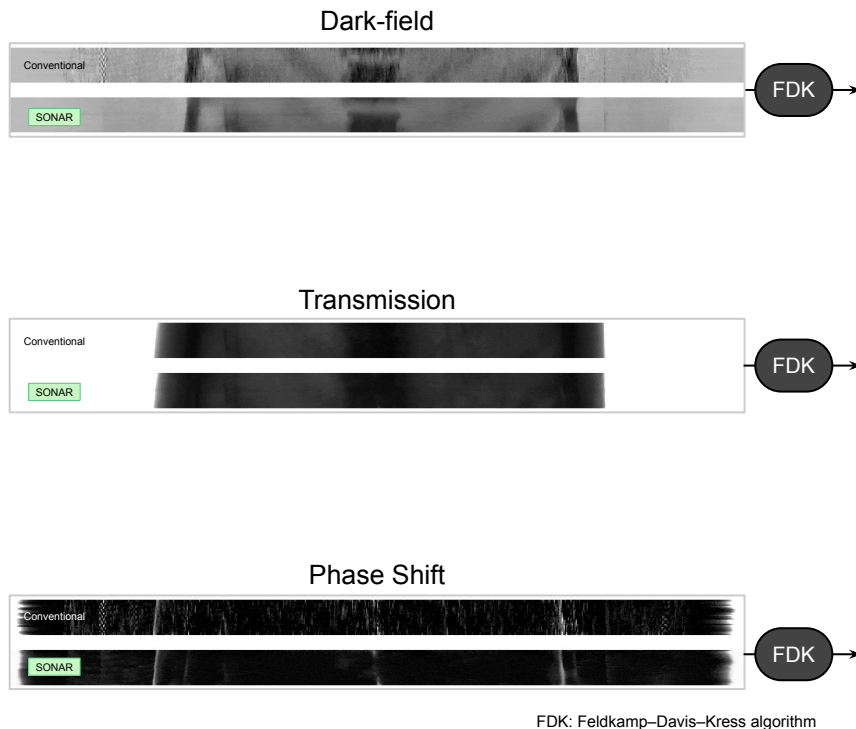
Transmission



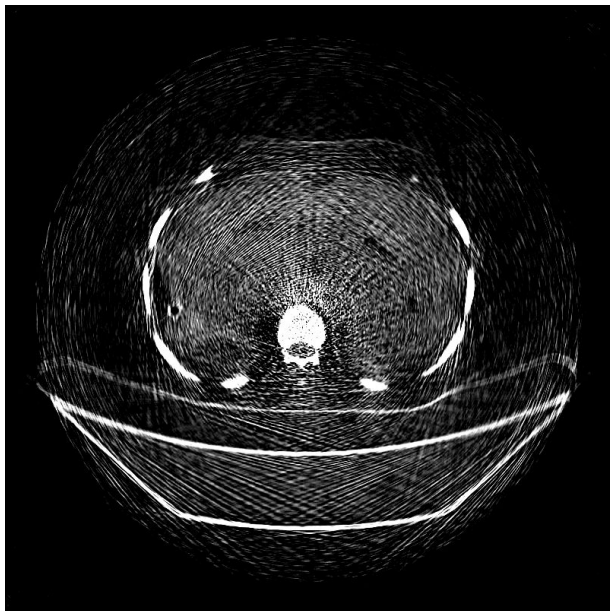
Phase Shift



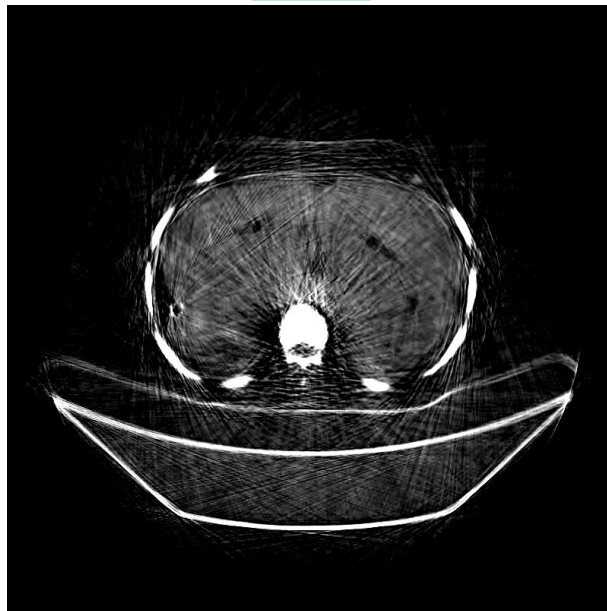
Reconstruction Space



Conventional



SONAR



Ground Truth



DFCT Metrics: Conventional $\rightarrow$ SONAR

PSNR_{3D} ($\uparrow$): 34.8 dB $\rightarrow$ 38.5 dB (+11%)

SSIM_{3D} ($\uparrow$): 0.831 $\rightarrow$ 0.938 (+13%)

Streak-reduced Human-scale Dark-field CT

with 3D Gaussian Splatting

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INTRODUCTION

- The clinical dark-field CT scanner (DFCT) enables attenuation- and small-angle scatter-based imaging of human-scale phantoms.
- Due to the scanner's vibrating Talbot-Lau grating interferometer, images suffer from inherent streak artifacts [1].
- In this work, DFCT data is processed by re-purposing the 3DGS Splatting (3DGS) method, designed for rendering sparsely sampled X-ray attenuation data [2], and is compared against conventional filtering:

- Gaussian filter ($\sigma = 1.00$)
- Bilateral filter ($C_{\text{Gauss}} = 5.60$)
- Total Variation (TV) filter ($\lambda = 0.04$)

RESULTS

The figure displays five columns of medical scan slices: Reference, Gaussian Filter, Bilateral Filter, TV Filter, and Gaussian Splatting. Each column shows two rows of images: one with a red box highlighting a specific area and another showing the full slice. Below the images is a line graph titled 'Mean ± Standard Deviation' comparing two metrics across the five methods: Contrast-to-noise Ratio (CNR) ↑ (blue line with circles) and Full width-at-half-maximum (FWHM) ↓ (orange line with squares). The x-axis labels are Reference, Gaussian, Bilateral, TV, and 3DGS. The y-axis ranges from 0 to 50. The graph shows that while all methods improve CNR, 3DGS achieves the highest CNR (~42), whereas the Gaussian filter maintains the lowest FWHM (~16).

Method	CNR _{FV} (↑)	CNR _{DOS} (↑)	FWHM _{DOS} (↓)
Reference	~2.5	~42.0	~42.0
Gaussian	~4.0	~32.0	~16.21
Bilateral	~6.0	~34.0	~16.21
TV	~7.0	~36.0	~16.21
3DGS	~42.0	~34.0	~16.21

CONCLUSION

These findings indicate that 3DGS outperforms conventional filtering in DFCT streak-reduction, preserving edge sharpness and a high CNR, a necessity for diagnostic accuracy in future clinical trials.

REFERENCES

- [1] M. Vermetz et al., "Dark-field computed tomography reaches the human scale," *PNAS*, vol. 119, 2022.
- [2] R. Zhu et al., "W-Gaussian: Redefining radiative gaussian splatting for tomographic reconstruction," *Advances in Neural Information Processing Systems*, vol. 37, 2024.

Acknowledgements

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Thank you!

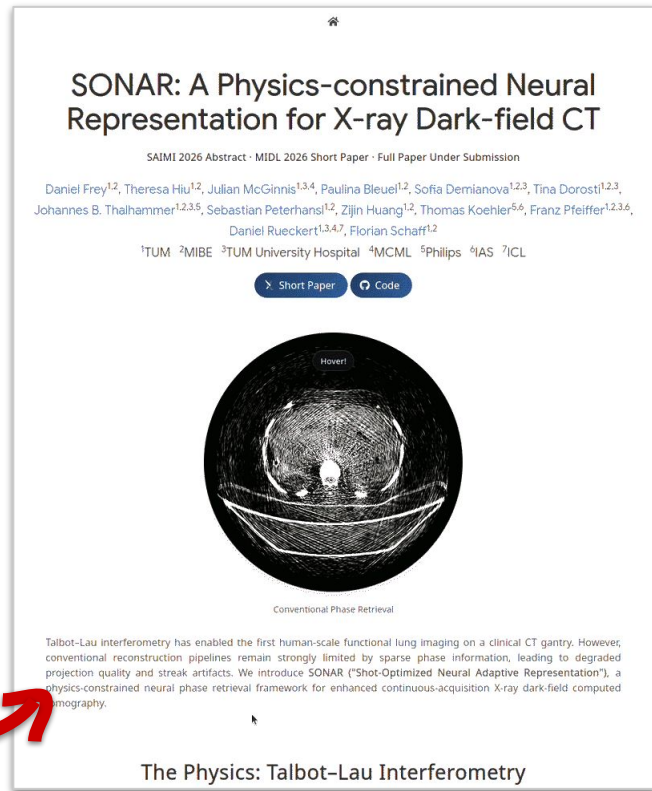
Project webpage:

[dcfrey.github.io/sonar](https://github.com/dcfrey.github.io/sonar)

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